Darke Theory Field Equation and the Prediction of Finite-Size Post-Threshold Matter

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Abstract

A standalone prediction is extracted from the Darke Theory Field Equation. The result is not merely that a nontrivial branch exists, but that any nontrivial finite-energy stationary matter sector generated by the equation cannot be truly pointlike. The static energy of the frozen parent theory splits into a quadratic current contribution and a quartic commutator contribution with opposite scaling under uniform spatial rescaling. This immediately forbids both zero-size collapse and infinite dispersion for any nontrivial finite-energy stationary branch. Any such branch must therefore occur at a finite preferred scale. The solved charge-one branch of the Darke Theory Field Equation is then shown to satisfy the corresponding virial balance to high numerical accuracy. The resulting prediction is exact in structural form and concrete in branch realisation: stable closure-bearing matter in this theory is intrinsically extended, possesses a nonzero characteristic radius, and must therefore exhibit a nontrivial form factor. This statement is independent of final particle identification and stands as a direct physical prediction of the field equation itself.

1. Introduction

The importance of a field equation is not exhausted by the existence of a solution. A physically meaningful field equation should also generate predictions about the kind of structure that can persist within its theory space. The present paper isolates one such prediction from the Darke Theory Field Equation.

The core point is simple. The equation does not merely admit a nontrivial localised branch. Its parent-action structure forces a stronger conclusion: any nontrivial finite-energy stationary post-threshold matter sector must occur at a finite preferred scale. Collapse to exact pointlikeness is forbidden by the energy balance of the theory itself. Likewise, unlimited dispersion is forbidden. A stationary matter sector in this theory is therefore not an exact point excitation but an intrinsically extended closure object.

This matters because it is a true statement piece. It does not depend on rhetorical amplification, and it does not require final empirical identification to be meaningful. It follows directly from the structure of the frozen parent action and is then realised explicitly in the solved charge-one branch.

The larger Darke Theory architecture gives this result additional significance. In that corpus, post-threshold geometry is the regime in which closure becomes geometrically encoded, and the Higgs is positioned as the first scalar geometric expression of closure. The present paper does not claim that the final particle identification step has been completed. It does claim that the field equation already forces a hard prediction about the character of any stable closure-bearing matter sector it generates: it must be finite-size and therefore cannot be truly pointlike.

The paper proceeds in a direct sequence. First, the static energy of the Darke Theory Field Equation is split into its quadratic and quartic sectors. Secondly, the scaling law under uniform spatial rescaling is derived. Thirdly, the finite-size theorem is proved. Fourthly, the exact virial identity for stationary branches is stated. Fifthly, the solved charge-one branch is checked numerically against that identity. Finally, the physical prediction is stated in its strongest clean form.

2. Static energy of the Darke Theory Field Equation

The Darke Theory Field Equation is frozen as a three-dimensional parent-action theory whose hedgehog reduction yields the static energy functional

$$E[F] = 4\pi \int_0^\infty \left[R^2 F'^2 + 2\sin^2 F + 2\sin^2 F\, F'^2 + \frac{\sin^4 F}{R^2} \right] dR. \tag{1}$$

For the present result, the key move is to separate this energy into two structural pieces:

$$E[F] = E_2[F] + E_4[F], \tag{2}$$

with

$$E_2[F] = 4\pi \int_0^\infty \left(R^2 F'^2 + 2\sin^2 F \right) dR, \tag{3}$$

and

$$E_4[F] = 4\pi \int_0^\infty \left(2\sin^2 F\, F'^2 + \frac{\sin^4 F}{R^2} \right) dR. \tag{4}$$

The term E_2 is the quadratic current sector. The term E_4 is the quartic commutator sector. The theorem below follows from the fact that these two pieces scale oppositely under uniform spatial rescaling.

For nontrivial finite-energy hedgehog branches, both sectors are strictly positive. Their integrands are nonnegative and are not identically zero for any nonconstant branch

satisfying the usual topological boundary conditions. The positivity assumption used below is therefore natural for the nontrivial sector under discussion.

3. Scaling law

Consider the rescaled profile family

$$F_\lambda(R) = F(\lambda R), \qquad \lambda > 0. \tag{5}$$

Lemma 1 (Scaling law)**.** *For the rescaled profile family* (5)*, the two energy pieces scale as*

$$E_2[F_\lambda] = \lambda^{-1} E_2[F], \tag{6}$$

and

$$E_4[F_\lambda] = \lambda E_4[F]. \tag{7}$$

Hence the full rescaled energy is

$$E(\lambda) = E[F_\lambda] = \lambda^{-1} E_2 + \lambda E_4. \tag{8}$$

Proof. For the quadratic piece,

$$E_2[F_\lambda] = 4\pi \int_0^\infty \left(R^2 (F_\lambda')^2 + 2\sin^2 F_\lambda \right) dR.$$

Since

$$F_\lambda'(R) = \lambda F'(\lambda R),$$

we obtain

$$E_2[F_\lambda] = 4\pi \int_0^\infty \left(R^2 \lambda^2 F'(\lambda R)^2 + 2\sin^2 F(\lambda R) \right) dR.$$

Set $s = \lambda R$, so that $R = s/\lambda$ and $dR = ds/\lambda$. Then

$$E_2[F_\lambda] = 4\pi\lambda^{-1} \int_0^\infty \left(s^2 F'(s)^2 + 2\sin^2 F(s) \right) ds = \lambda^{-1} E_2[F].$$

Likewise, for the quartic piece,

$$E_4[F_\lambda] = 4\pi \int_0^\infty \left(2\sin^2 F_\lambda (F_\lambda')^2 + \frac{\sin^4 F_\lambda}{R^2} \right) dR.$$

Substituting $F_\lambda(R) = F(\lambda R)$, $F_\lambda'(R) = \lambda F'(\lambda R)$, and again using $s = \lambda R$, gives

$$E_4[F_\lambda] = 4\pi\lambda \int_0^\infty \left(2\sin^2 F(s)\, F'(s)^2 + \frac{\sin^4 F(s)}{s^2} \right) ds = \lambda E_4[F].$$

So the two sectors scale oppositely, and (8) follows immediately. □

4. Finite-Size Theorem

Theorem 1 (Finite-Size Theorem for the Darke Theory Field Equation)**.** *Let F be any nontrivial finite-energy static configuration of the Darke Theory Field Equation with*

$$E_2[F] > 0, \qquad E_4[F] > 0. \tag{9}$$

Then the theory forbids both collapse to zero size and dispersion to infinite size. Any stationary nontrivial branch must therefore occur at a finite preferred scale.

Proof. By the scaling law,

$$E(\lambda) = \lambda^{-1}E_2 + \lambda E_4.$$

As $\lambda \to 0$,

$$E(\lambda) \sim \lambda^{-1}E_2 \to \infty,$$

since $E_2 > 0$. So zero-size collapse is forbidden.

As $\lambda \to \infty$,

$$E(\lambda) \sim \lambda E_4 \to \infty,$$

since $E_4 > 0$. So infinite spreading is forbidden.

Therefore any nontrivial stationary branch must occur at some finite scale λ_*. Differentiating,

$$\frac{dE}{d\lambda} = -\lambda^{-2}E_2 + E_4. \tag{10}$$

Setting (10) equal to zero gives

$$\lambda_* = \sqrt{\frac{E_2}{E_4}}. \tag{11}$$

This is finite because both sectors are strictly positive.

The second derivative is

$$\frac{d^2E}{d\lambda^2} = 2\lambda^{-3}E_2 > 0, \tag{12}$$

so the stationary point is stable against uniform rescaling.

Thus any nontrivial stationary branch of the Darke Theory Field Equation must have finite preferred size. □

5. Virial identity

Corollary 1 (Virial identity for an exact stationary branch)**.** *Let F be an exact finite-energy stationary branch of the Darke Theory Field Equation. Then*

$$E_2[F] = E_4[F]. \tag{13}$$

Proof. If F is an exact stationary branch, then $\lambda = 1$ is stationary along the rescaled family $F_\lambda(R) = F(\lambda R)$. Therefore

$$\left.\frac{dE}{d\lambda}\right|_{\lambda=1} = 0.$$

Using (10), this gives

$$-E_2 + E_4 = 0,$$

hence (13). □

This is the exact balance condition for an exact stationary branch. It is not optional and not interpretive. It is the direct signature of finite-size balance in the parent-action theory.

It is important to note the scope precisely. The finite-size theorem and virial identity control the behaviour of the branch under uniform rescaling. They therefore prove finite preferred scale and exclude pointlike collapse and infinite dispersion. They do not by themselves establish full perturbative stability in every fluctuation channel. That stronger claim belongs to the broader classical audit structure of the field-equation paper.

6. Numerical branch check

The solved charge-one branch of the Darke Theory Field Equation gives the total dimensionless energy

$$E \approx 145.85782948. \tag{14}$$

Splitting that branch numerically into the quadratic and quartic pieces gives

$$E_2 \approx 72.91205346, \tag{15}$$

and

$$E_4 \approx 72.94577602. \tag{16}$$

So

$$E_2 + E_4 \approx 145.85782948, \tag{17}$$

in agreement with the solved total branch energy, and

$$E_2 - E_4 \approx -0.03372256. \tag{18}$$

Equivalently,

$$\frac{E_2 - E_4}{E_2 + E_4} \approx -2.31 \times 10^{-4}. \tag{19}$$

This verifies that the solved charge-one branch sits essentially exactly at the finite-size virial balance point. The theorem is therefore not merely a formal scaling statement about the action in general. It is realised directly by the actual live branch of the theory.

7. Immediate physical consequences

The theorem and virial check together yield several immediate consequences.

First, the stationary nontrivial sector cannot be truly pointlike. Exact collapse is forbidden by the divergence of the quadratic energy sector under rescaling.

Secondly, the sector cannot be infinitely diffuse. Unlimited spreading is forbidden by the divergence of the quartic sector.

Thirdly, a finite intrinsic size is not a contingent property of one branch among many. It is enforced by the structure of the field equation itself.

Fourthly, once a finite nonzero spatial extent exists, a nontrivial form factor becomes unavoidable. A truly structureless exact point excitation would not carry such internal scale. The solved branch of the Darke Theory Field Equation already exhibits explicit dimensionless size and response markers, including a half-energy radius, a root-mean-square radius, and a half-form-factor scale.

The prediction therefore takes its strongest clean form as follows.

> Any stable nontrivial post-threshold matter sector generated by the Darke Theory Field Equation is an intrinsically extended topological object with nonzero characteristic size and a nontrivial form factor, rather than a truly pointlike elementary excitation.

Within the broader interpretive architecture of Darke Theory, this applies in particular to the first stable closure-bearing matter sector. That is the architectural significance of the present theorem-level result.

8. Relation to closure structure

This prediction matters not only because it follows from the frozen parent action, but because of where the field equation sits inside the larger Darke Theory architecture.

In the wider corpus, post-threshold geometry is the regime in which closure becomes geometrically encoded. Within that same architecture, the Higgs is positioned as the first scalar geometric expression of closure. The present result does not assert final identification of the solved branch with that empirical sector. It does establish the first hard structural prediction one would expect if the broader architecture is correct: the first stable closure-bearing matter sector cannot be truly pointlike.

That is the exact strength of the present paper. It does not claim more than has been earned. It also does not retreat from the force of what has been earned. The prediction is direct, formal, and branch-realised.

9. Boundary of the claim

This paper does not claim exhaustive control of every coupled nonradial sector. It does not claim full phenomenological closure. It does not claim quantum completion or final particle identification. Those remain open.

What it does claim is exact. The scaling structure of the Darke Theory Field Equation forces any nontrivial finite-energy stationary matter sector to occur at finite preferred scale. The solved charge-one branch realises that balance to high numerical accuracy. Therefore finite-size post-threshold matter is not a speculative possibility within the theory. It is a direct prediction of the field equation.

10. Conclusion

A genuine prediction has now been extracted from the Darke Theory Field Equation.

The theorem is simple and hard. Because the quadratic and quartic energy sectors scale oppositely under uniform spatial rescaling, collapse and infinite spreading are both forbidden for any nontrivial finite-energy stationary branch. Exact stationary branches therefore satisfy a virial balance between the two sectors. The solved charge-one branch satisfies that balance to high precision. The resulting physical statement is unavoidable: post-threshold matter in this theory is intrinsically extended.

This is not merely the claim that a localised branch exists. It is the stronger claim that the field equation itself forbids true pointlikeness for any nontrivial finite-energy stationary matter sector it generates. In the broader architecture of Darke Theory, that result applies directly to the first stable closure-bearing matter sector. The statement stands independently of final particle identification and belongs to the direct physical content of the theory.

References

References

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